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A cross-temporal contrastive disentangled model for ancient Chinese understanding

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ABSTRACT

Ancient Chinese is a crucial bridge for understanding Chinese history and culture. Most existing works utilize high-resource modern Chinese to understand low-resource ancient Chinese, but they fail to fully consider the semantic and syntactic gaps between them due to their changes over time, resulting in the misunderstanding of ancient Chinese. Hence, we propose a novel language pre-training framework for ancient Chinese understanding based on the Cross-temporal Contrastive Disentanglement Model (CCDM), which bridges the gap between modern and ancient Chinese with their parallel corpus. Specifically, we first explore a cross-temporal data augmentation method by disentangling and reconstructing the parallel ancient-modern corpus. It is noteworthy that the proposed decoupling strategy takes full account of the cross-temporal character between ancient and modern Chinese. Then, cross-temporal contrastive learning is exploited to train the model by fully leveraging the cross-temporal information. Finally, the trained language model is utilized for downstream tasks. We conduct extensive experiments on six ancient Chinese understanding tasks. Results demonstrate that our model outperforms the state-of-the-art baselines. Our framework also holds potential applicability to other languages that have undergone evolutionary changes, leading to shifts in syntax and semantics.¹

1. Introduction

Ancient Chinese texts are concise and obscure, but record a long history of more than 5,000 years, providing an important bridge to understanding ancient Chinese culture. Compared with modern Chinese, ancient Chinese has not been fully digitized and the corpus is relatively less. As a result, the understanding of ancient Chinese has garnered interest from specialists across various fields such as natural language processing (NLP) and computational linguistics (Shu, Guo, Wang, Zhang, & Hu, 2021).

In the early period, one of the methods to comprehend ancient texts was manual understanding (Liu, Wu, & Bai, 2020), which includes the usage of modern Chinese annotations provided by experts, or consulting ancient Chinese dictionaries and matching them with reasonable interpretations. The first approach requires extensive expert resources, while the second needs reasonable judgment. All these approaches are time-consuming and labor-intensive.

Considering the limitations of manual understanding, the development of automated language understanding models utilizing pre-trained language models (PLMs) has emerged as a solution (Devlin, Chang, Lee, & Toutanova, 2019). Most of these methods were trained on modern Chinese (Cui, Che, Liu, Qin, & Yang, 2021; Sun et al.,

2021), which are not completely suitable for ancient Chinese. This is because, as time progresses, modern Chinese have changed to some extent compared to ancient Chinese. For example, in Fig. 1(a), the underlined word “与” (yǔ) means “compare with” in ancient Chinese, but more often means “with” in modern Chinese. Hence, some researchers have fine-tuned existing Chinese PLMs on the ancient Chinese corpora and made some progress in ancient Chinese understanding, such as AnchiBERT (Tian, Yang, Liu, & Lv, 2021), SikuBERT (Wang et al., 2022) and GujiBERT (Wang, Liu, et al., 2023). However, these methods focused more on semantic changes and did not effectively capture syntactic variations (Wang, Zhang, Li, & Hou, 2023) (as illustrated in Fig. 1(b), where the syntax of sentences displays **partial** divergence). The motivation for this paper is how to draw on the cross-temporal property, i.e., the similarities and differences in the semantics and the syntax between modern Chinese and ancient Chinese, so as to assist in the understanding of ancient Chinese.

To overcome the limitations of existing methods, this paper proposes a novel language pre-training framework with the Cross-temporal Contrastive Disentanglement Model (CCDM) for ancient Chinese understanding, which breaks the gap between modern and ancient Chinese

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¹ Our model is available at <https://github.com/yuting-wei/CCDM>.

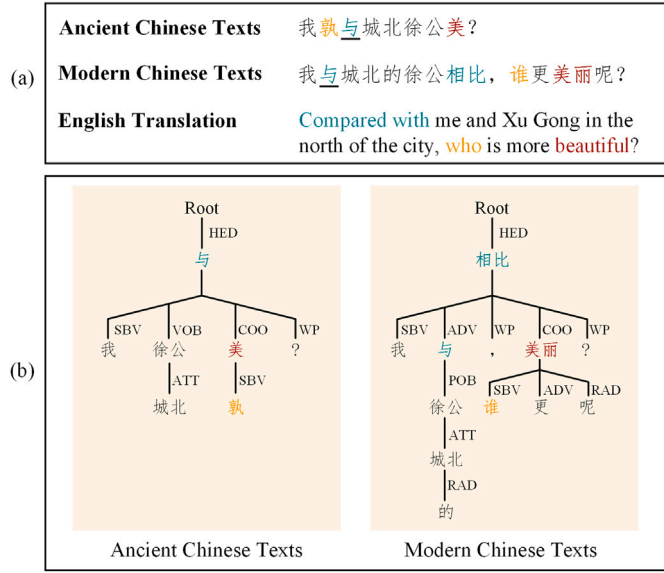


Fig. 1. A parallel sentence pair for ancient and modern Chinese. The words in the same color express the same semantics. (a) Parallel sentences and their English translation. (b) Dependency tree of parallel sentences.

with limited parallel corpus. Specifically, CCDM consists of two components: cross-temporal data augmentation and cross-temporal contrastive learning. First, a cross-temporal data augmentation method is designed by disentangling the parallel corpus (ancient-modern) and reconstructing the cross-temporal sentence representation. In particular, inspired by Chen, Tang, Wiseman, and Gimpel (2019), we propose a novel disentanglement strategy, i.e., VG²VAE, which decouples the text representation into three components, including semantics, the same syntax, and different syntax. Differing from the previous strategy (Chen et al., 2019), this approach takes into account the similarities and differences in syntax between ancient and modern Chinese (as shown in Fig. 1(b)). Second, cross-temporal contrastive learning is proposed to train the PLM and enhance the effect of decoupling. Concretely, the reconstructed original ancient sentence representation is chosen as the anchor. Positives are the cross-temporal reconstructed representations of ancient sentences, while negatives are the cross-temporal reconstructed representations of modern sentences. The PLM, trained through contrastive learning, learns the similarities and differences in representational space between ancient and modern Chinese, especially in syntax. Finally, the trained language model (PLM) is utilized for downstream tasks.

The contributions of this paper are as follows.

- To our best knowledge, this is the first effort to explore the cross-temporal property with Cross-temporal Contrastive Disentangling Model (CCDM) for ancient Chinese understanding (ACU), providing a new perspective for bridging the syntax gap between modern and ancient Chinese.
- A novel decoupling strategy for semantics and syntax is proposed, i.e., VG²VAE, which takes into account syntactic similarities and facilitates the utilization of cross-temporal information.
- Extensive experiments are conducted on 6 ACU tasks, achieving better performance than the state-of-the-art baselines.

2. Related work

2.1. PLMs for Chinese

PLMs have taken natural language processing into a new era, attracting a growing number of researchers in different languages (Devlin

et al., 2019; Ouyang et al., 2021). The aim of the recent PLM is to learn dynamic word representation. For example, Bi-LSTM-based methods such as ELMo (Peters et al., 2018), as well as transformer-based methods such as BERT (Devlin et al., 2019) and ERNIE (Zhang et al., 2019). In particular, considering the specificity of Chinese word segmentation and glyph as well as pinyin, several related methods are proposed (Cui et al., 2021; Sun et al., 2021). For instance, considering that the word in Chinese is a combined unit rather than an independent unit in English, Cui et al. (2021) introduced the concept of Whole Word Masking, an approach that involved masking all Chinese characters constituting a single word. Sun et al. (2021) developed ChineseBERT, which integrated both glyph and pinyin information of Chinese characters during its pre-training phase.

However, most of these Chinese PLMs are trained on corpora biased to modern Chinese, which are not well applicable to the language understanding of ancient Chinese. Consequently, some scholars have tried to fine-tune the existing Chinese PLMs on the ancient Chinese corpus to enable the models to perform well on the ancient Chinese understanding tasks (Tian et al., 2021; Wang et al., 2022). For example, AnchiBERT (Tian et al., 2021) trained the PLM on ancient Chinese texts from 1000 B.C. to 200 B.C.. SikuBERT (Wang et al., 2022) constructed a high-quality corpus of ancient texts based on the *Si Ku Quan Shu*² and trained language models on this corpus. However, fine-tuning is susceptible to corpus size, which leads to a decrease in the generalization ability of PLMs.

Several large language models (LLMs) have been developed specifically for understanding Chinese language (Yang et al., 2023; Zeng et al., 2022). These models have shown promising results in various applications. Despite this, they are particularly adept at general language understanding and require additional fine-tuning for specialized downstream tasks, necessitating substantial computational resources. Consequently, this paper focuses on the study of smaller, more easily deployable pre-trained models.

2.2. PLMs for low-resource languages

In the context of languages with limited resources, it is more beneficial to utilize data from higher-resource languages that exhibit linguistic similarities, particularly when a substantial lexical overlap exists (Hedderich, Lange, Adel, Strötgen, & Klakow, 2021). There are two common approaches for pre-training low-resource languages: data augmentation (Chen, Tam, Raffel, Bansal, & Yang, 2023; Mi, Xie, & Zhang, 2022; Zhang, Yang, & Yang, 2022) and cross-language transfer (Conneau et al., 2020; Luo et al., 2021; Ouyang et al., 2021).

Most of the existing data augmentation methods, such as random word insertion or deletion, may destroy the syntactic structure within the language, which is not conducive to language understanding. For cross-lingual transfer, parallel corpora play an instrumental role, especially when enhancing the semantic comprehension of low-resource languages through high-resource languages (Conneau et al., 2020; Ouyang et al., 2021; Xu et al., 2023; Zhang, Tan, Huang, & Huang, 2023). For example, Conneau et al. (2020) proposed a translation language model (TLM) based on MLM, which is designed to effectively train a model's cross-lingual capabilities by the cross-lingual parallel corpora. Compared with MLM which only considered monolingual input sequences, TLM simultaneously concatenated parallel bilingual sequences as inputs. Ouyang et al. (2021) introduced two novel approaches called Cross-attention Masked Language Model and Back-translation Masked Language Model to improve multilingual representation by aligning the semantic understanding of different languages with monolingual datasets. Considering the high resource nature of modern Chinese, these methods provide a road to ancient Chinese

² The *Si Ku Quan Shu* contains valuable cultural books from the pre-Qin dynasty down to the Qing dynasty, covering more than 2,000 years.

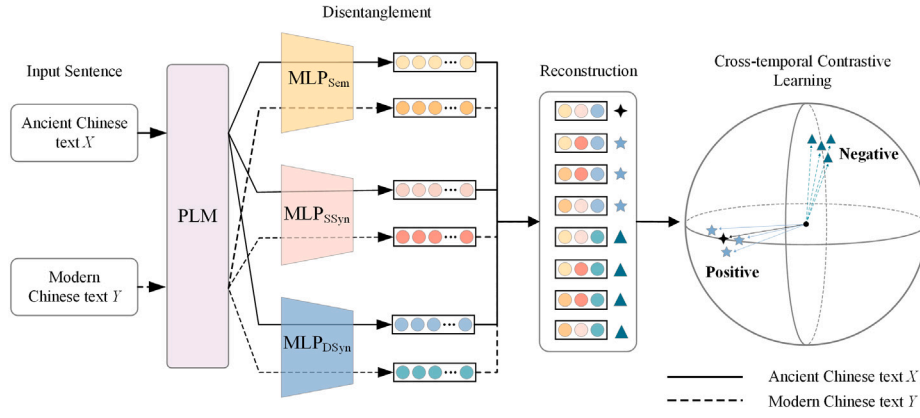


Fig. 2. An overview of our proposed CCDDM framework. Given a parallel sentence pair, we first extract their representations utilizing a shared fine-tunable pre-trained model. Then we introduce a dedicated disentanglement-reconstruction strategy to perform decoupling-based data augmentation by randomly combining decoupled semantic and syntactic information. Finally, we leverage cross-temporal contrastive learning to optimize the model by separately distinguishing and constraining the spatial representations of ancient and modern texts.

understanding. However, challenges arise from the scarcity of parallel corpora. Nevertheless, these low-resource pre-training methods emphasize learning language-agnostic semantic representations and ignore the importance of syntax for language understanding (Giulianelli, Kutuzov, & Pivovarov, 2022). Hence, the existing low-resource methods cannot solve this problem well.

2.3. Applications of disentangled representation learning

The concept of Disentangled Representation, initially introduced by Bengio (Bengio et al., 2013), focuses on learning low-dimensional and interpretable representations, which identify and separate different potentially changing factors within high-dimensional data. This approach has gained widespread usage in the field of computer vision (Li et al., 2019), for example, by separating image style and content in the latent space to achieve artistic style transfer (Zhang, Zhang, & Cai, 2018).

Recently, there has been an increasing number of studies on learning disentangled representation and its related applications in NLP tasks, such as sentence style transfer (Nangì, Chhaya, Khosla, Kaushik, & Nyati, 2021), text generation (Balasubramanian, Kobayev, Bahuleyan, Shapiro, & Vechtomova, 2021), and knowledge graph completion (Wu et al., 2021). The disentangled representation of text is based on variational autoencoders (VAE) (Xu & Durrett, 2018). One of the most relevant to us is the vmf-Gaussian Variational Autoencoder (VGVAE) model proposed by Chen et al. (2019), a work on syntax-semantics decoupling for monolingual texts. It uses two latent variables to represent the semantics and syntax of the reconstructed sentence, where the semantic representation is learned by paraphrase reconstruction loss and discriminative paraphrase loss, and the syntactic representation is learned by word-order information. The difference in our work is that our texts are cross-temporal. In order to better learn the gap between ancient and modern Chinese, we propose a novel decoupling-based cross-temporal text augmentation strategy, which sufficiently takes into account the fine-grained gap in semantics and syntax.

3. Problem formulation

Pre-training language models (PLMs) are developed to exploit the underlying knowledge of massive unlabeled texts to understand language automatically. For ancient Chinese understanding (ACU), some scholars have improved these PLMs by fine-tuning them on ancient corpora. However, limited by the scarcity of ancient texts, it is not enough for these PLMs to sufficiently learn the difference between ancient and modern Chinese.

In contrast to previous work that simply fine-tuned these PLMs, we propose a novel language pre-training framework for ACU from the perspective of the semantic and syntactic gap between ancient and modern Chinese. The overall architecture of the language framework is illustrated in Fig. 2. Given a number of parallel sentence pairs (x, y) , where $x = \{x_1, x_2, \dots, x_n\}$ is the ancient Chinese text and $y = \{y_1, y_2, \dots, y_l\}$ is the modern Chinese text, our goal is to fine-tune the PLM by the cross-temporal contrastive learning, which fully leverages the cross-temporal property. Then, the trained language model (PLM) is utilized for downstream ACU tasks.

4. CCDDM framework

The full pipeline of the proposed CCDDM framework is shown in Fig. 2, which explores the cross-temporal property with parallel sentence pairs. In our framework, there are three important components: (1) input sentence representation layer (to get word embeddings of ancient Chinese text x and modern Chinese text y), (2) disentanglement (to decouple the sentence embedding into three parts: semantic, same syntax, and different syntax) and reconstruction layer (to reconstruct the original and cross-temporal sentence representations by randomly selecting the semantic and syntactic variables), (3) cross-temporal contrastive learning (to train the PLM and disentanglement layer). In the following subsections, we detail the language framework.

4.1. Input sentence representation layer

For each parallel sentence pair, a shared fine-tunable pre-trained model RoBERTa-wwm (Cui et al., 2021) is utilized to obtain word embeddings for the ancient Chinese sentence x and the modern Chinese sentence y .

For an ancient Chinese sentence $x = \{x_1, x_2, \dots, x_n\}$, each x_i corresponds to the i th word in the sentence x , with n representing the sentence's length. Then, we employ the fine-tunable pre-trained model RoBERTa-wwm (Cui et al., 2021) for learning word embeddings and obtain the representation $H^x = \{h_1^x, h_2^x, \dots, h_n^x\}$ of the sentence x .

Given a modern Chinese sentence $y = \{y_1, y_2, \dots, y_l\}$, each y_j represent j th word of the sentence y and l is the length of the sentence y . Then, we learn the word embeddings with the same fine-tunable pre-trained model RoBERTa-wwm (Cui et al., 2021) and obtain the representation $H^y = \{h_1^y, h_2^y, \dots, h_l^y\}$ of the sentence y .

4.2. Disentanglement and reconstruction layer

The motivation for disentanglement and reconstruction is rooted in the need to effectively harness the capabilities of robust pre-trained models developed for modern Chinese, thereby facilitating transfer learning to ancient Chinese texts. Traditionally, parallel corpora have been instrumental in cross-linguistic transfer tasks. Prior methodologies, such as the von Mises-Fisher (vMF) Gaussian Variational Autoencoder (VGVAE) presented by Chen et al. (2019), primarily decompose sentence representations into semantic and syntactic components for training purposes. However, these approaches generally neglect the nuanced syntactic similarities shared between ancient and modern Chinese, which share a common lineage but partly differ in usage and syntax over time.

To address this challenge, we propose an enhanced model, the VG²VAE, as illustrated in Fig. 2. This model advances beyond the conventional VGVAE by assuming that the generation of a sentence is influenced by three independent variables: a semantic variable (c), a same syntactic variable (s_1), and a different syntactic variable (s_2). This tripartite variable framework is designed to better capture the linguistic continuum between ancient and modern Chinese, enabling more effective transfer learning.

Take the ancient Chinese sentence x as an example. It can be decoupled into

$$c^x, s_1^x, s_2^x = D(H^x), \quad (1)$$

where D is a disentangled separator based on VG²VAE.

Formally, the joint probability of the sentence x and its three latent variables can be factorized as:

$$p_\theta(x, c^x, s_1^x, s_2^x) = p_\theta(c)p_\theta(s_1)p_\theta(s_2)p_\theta(x | c^x, s_1^x, s_2^x), \quad (2)$$

where $p_\theta(x | c^x, s_1^x, s_2^x)$ is a generative model consisting of bag-of-words decoder for reconstructing the sentence x .

$$p_\theta(x | c^x, s_1^x, s_2^x) = \prod_{i=1}^n p(x_i | c^x, s_1^x, s_2^x), \quad (3)$$

where $p(x_i | c^x, s_1^x, s_2^x)$ is determined through a softmax function applied over a vocabulary consisting of V words.

The variational inference process of VG²VAE uses a factorized approximated posterior $q_\phi(c^x | x)q_\phi(s_1^x | x)q_\phi(s_2^x | x) = q_\phi(c^x, s_1^x, s_2^x | x)$ with the objective function that maximizes a lower bound of the marginal log-likelihood:

$$\begin{aligned} \mathcal{L}_{VG^2VAE}^X &= \mathcal{L}_{RL}^X + \text{KL}(q_\phi(c^x | x) \| p_\theta(c^x)) \\ &\quad + \text{KL}(q_\phi(s_1^x | x) \| p_\theta(s_1^x)) \\ &\quad + \text{KL}(q_\phi(s_2^x | x) \| p_\theta(s_2^x)), \end{aligned} \quad (4)$$

$$\mathcal{L}_{RL}^X = \mathbb{E}_{\substack{c^x \sim q_\phi(c^x | x) \\ s_1^x \sim q_\phi(s_1^x | x) \\ s_2^x \sim q_\phi(s_2^x | x)}} [-\log p_\theta(x | c^x, s_1^x, s_2^x)], \quad (5)$$

where $q_\phi(c^x | x)$ is subject to vMF($\mu_\alpha(x), \kappa_\alpha(x)$), while $q_\phi(s_1^x | x)$ and $q_\phi(s_2^x | x)$ respectively follow $N(\mu_\beta(x), \text{diag}(\kappa_\beta(x)))$ and $N(\mu_\gamma(x), \text{diag}(\kappa_\gamma(x)))$. The prior $p_\theta(c^x)$ follows the uniform distribution vMF($\cdot, 0$). The prior $p_\theta(s_1^x)$ and $p_\theta(s_2^x)$ follow the standard Gaussian distribution. We adopt multi-layer perception (MLP) networks to learn the mean (μ) and variance (κ) of three distributions. In particular, the parameters of three MLP networks are shared between the ancient and modern Chinese. Specifically, MLP_{Sem} , MLP_{SSyn} and MLP_{DSyn} in Fig. 2 are for semantics, same syntax and different syntactic, respectively. Eq. (5) is the reconstruction loss (RL) for the generator, which is computed by a bag-of-words decoder. $\mathcal{L}_{VG^2VAE}^Y$ is similar to $\mathcal{L}_{VG^2VAE}^X$.

Moreover, word position information is useful for disassociating syntactic information into syntactic latent variables (Chen et al., 2019). Since the bag-of-words decoder lacks sequential data, this information

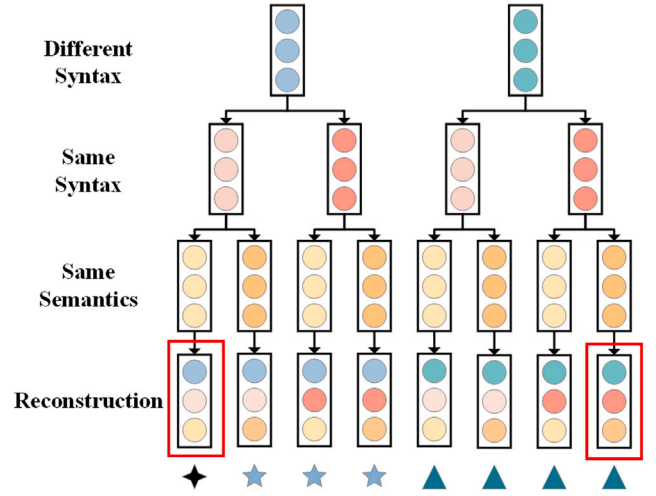


Fig. 3. Visualization of data augmentation through reconstruction. The left four reconstructed representations denote ancient texts, and the right four represent modern texts. Representations in red boxes are original reconstructions, while others are cross-temporal reconstructions.

can be used to compensate for this deficit. To utilize word order information, the word position loss (WPL) is defined as

$$\mathcal{L}_{WPL} = \mathcal{L}_{WPL}^X + \mathcal{L}_{WPL}^Y, \quad (6)$$

$$\mathcal{L}_{WPL}^X = \mathbb{E}_{\substack{s_1^x \sim q_\phi(s_1^x | x) \\ s_2^x \sim q_\phi(s_2^x | x)}} \left[- \sum_i \log \text{softmax}(f([h_i^x; s_1^x; s_2^x]))_i \right], \quad (7)$$

where $\text{softmax}(\cdot)_i$ denotes the probability of the i th word occurring at position i , and $f(\cdot)$ represents a three-layer feed-forward neural network. The input to $f(\cdot)$ is derived from the concatenation of the same syntactic variable s_1^x , the different syntactic variable s_2^x and the embedding h_i^x of the i th token. $\mathcal{L}_{WPL}^Y$ is similar to $\mathcal{L}_{WPL}^X$.

After decoupling, we conduct data augmentation through a reconstruction process, as illustrated in Fig. 3. We reconstruct the sentence representations by randomly selecting the semantic and syntactic variables. The style of the reconstructed sentence is determined by s_2 , since it is assumed that both c and s_1 contain the same information regarding parallel sentences. For example, a sentence representation can be obtained by concatenating the c^y , s_1^y , s_2^x together, which expresses the semantics of the modern Chinese sentence y in the syntactic form of the ancient Chinese sentence x .

4.3. Cross-temporal contrastive learning

In training our ancient Chinese pre-training model (PLM) and its disentanglement layer, we utilize contrastive learning to deepen the model's understanding of both the similarities and differences within parallel corpora from distinct historical periods. This training strategy is supported by two specifically designed loss functions: the cross-temporal contrastive loss and the cross-temporal reconstruction loss.

The cross-temporal contrastive loss utilizes a contrastive objective function enhanced with decoupling-based data augmentation. This function is designed to spatially differentiate the semantic domains of ancient and modern texts, facilitating proximity between texts (anchors and positive samples) that are from the same historical period and increasing the distance between the anchor and texts from different periods (negative samples), as depicted in Fig. 4. This method enhances the model's ability to distinguish texts across different epochs effectively.

The second loss, the cross-temporal reconstruction loss, enforces constraints on the reconstruction vectors of texts from the same period.

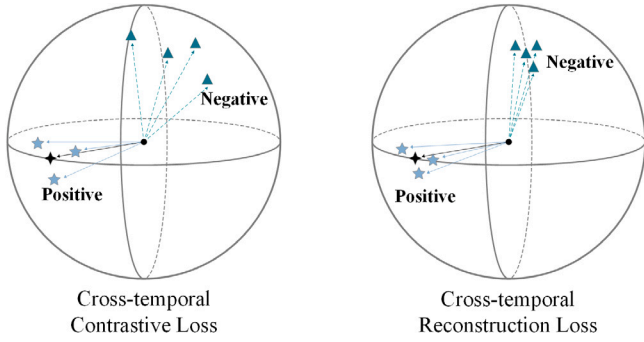


Fig. 4. An illustration of the difference between the two losses. The quadrangle represents the anchor, the pentagrams represent the positive samples, and the triangles represent the negative samples.

This loss function clusters both positive and negative samples with strong constraints, improving the disentanglement layer's ability to accurately model and reconstruct the linguistic features specific to each period. This structured approach not only supports better separation of textual features across different historical contexts but also enhances the overall interpretative capacity of the model.

Next, we detail the two losses.

4.3.1. Cross-temporal contrastive objective function

For each parallel sentence pair, we first obtain the positive and negative examples of ancient Chinese by reconstructing the sentence representations using the decoupled variables. There are 2^3 reconstructed sentence representations, $R = \{r_i; i = 1, \dots, 8\}$, as shown in Fig. 3. Specifically, the self-reconstructed representation of ancient text is regarded as the anchor, and the other reconstructed representations are divided into positive samples (P) and negative samples (N) according to the characteristics of their syntactic information (i.e., s_2). For example, the sentence representations composed of s_2^x are regarded as positive samples, since it is in the syntactic structure of the ancient Chinese. We define a cross-temporal contrastive objective function between these samples:

$$\mathcal{L}_{CCL} = -\log \frac{\sum_{i=2}^8 \exp(\delta(r_i) \text{sim}(r_1, r_i) / \tau)}{\sum_{i=2}^8 \exp((1 - \delta(r_i)) \text{sim}(r_1, r_i) / \tau)}, \quad (8)$$

$$\delta(r_i) = \begin{cases} 1 & r_i \in P \\ 0 & r_i \in N, \end{cases} \quad (9)$$

where $\delta(r_i)$ denotes whether r_i is a representation of ancient Chinese, r_1 represents the reconstructed representations of the ancient sentence x , τ denotes a temperature hyperparameter and $\text{sim}(\cdot)$ represents the cosine similarity function.

4.3.2. Cross-temporal reconstruction loss

Furthermore, we incorporate the cross-temporal reconstruction loss (CRL) as a strong constraint for the cluster of positive and negative samples. The foundational hypothesis underlying the CRL is that for a parallel sentence pair (x, y) , the semantic and same syntactic information are equivalent and only the different syntactic information varies.

We attempt to reconstruct the sentences by swapping the semantic and the same syntactic variables, while preserving the different syntactic variables. Specifically, the loss is defined as

$$\mathcal{L}_{CRL} = \mathcal{L}_{CRL}^X + \mathcal{L}_{CRL}^Y, \quad (10)$$

$$\begin{aligned} \mathcal{L}_{CRL}^X = & \mathbb{E}_{\substack{c^y \sim q_\phi(c^y|y) \\ s_1^x \sim q_\phi(s_1^x|x) \\ s_2^x \sim q_\phi(s_2^x|x)}} [-\log p_\theta(x | c^y, s_1^x, s_2^x)] \\ & + \mathbb{E}_{\substack{c^x \sim q_\phi(c^x|x) \\ s_1^y \sim q_\phi(s_1^y|y) \\ s_2^y \sim q_\phi(s_2^y|y)}} [-\log p_\theta(x | c^x, s_1^y, s_2^y)] \\ & + \mathbb{E}_{\substack{c^y \sim q_\phi(c^y|y) \\ s_1^y \sim q_\phi(s_1^y|y) \\ s_2^x \sim q_\phi(s_2^x|x)}} [-\log p_\theta(x | c^y, s_1^y, s_2^x)], \end{aligned} \quad (11)$$

$\mathcal{L}_{CRL}^Y$ is similar to $\mathcal{L}_{CRL}^X$. It is worth noting that two other reconstruction losses have been included in $\mathcal{L}_{VG^2VAE}^X$ and $\mathcal{L}_{VG^2VAE}^Y$.

The training objective of cross-temporal contrastive learning is defined as follows:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{VG^2VAE} + \lambda_2 \mathcal{L}_{WPL} + \lambda_3 \mathcal{L}_{CCL} + \lambda_4 \mathcal{L}_{CRL}, \quad (12)$$

where $\mathcal{L}_{VG^2VAE} = \mathcal{L}_{VG^2VAE}^X + \mathcal{L}_{VG^2VAE}^Y$ (see details in Section 4.2), $\mathcal{L}_{WPL}$ is the word position loss, $\mathcal{L}_{CCL}$ is the cross-temporal contrastive loss, $\mathcal{L}_{CRL}$ is the cross-temporal reconstruction loss and λ_i is the balancing term.

5. Experiments and results

This section is structured to first present the model settings and introduce a series of state-of-the-art baseline models for comparison. Subsequently, we proceed to evaluate the effectiveness of our CCDM across 6 ancient NLP tasks. Lastly, we conduct in a range of ablation studies aimed at assessing the impact of various components within our model.

5.1. Setup

We subsample 900K parallel sentence pairs from *Classical-Modern Corpus*³ as our training set, which is a relatively comprehensive parallel corpus of ancient and modern Chinese and covers most classic ancient Chinese books. We randomly use 72k parallel sentence pairs from Classical-Modern Corpus as a development set, which has no overlap with the training set. Consistent with the previous pre-trained models for ancient Chinese, we have also trained our language model using simplified Chinese characters.

For hyperparameters, our PLM has the same structure as RoBERTa-wm, consisting of 12 Transformer layers. Each layer has an input size of 768 and includes 12 heads. The sequence length is 128, since the input is a single sentence. The latent variables c , s_1 , and s_2 are each represented by a dimension of 256. The model has trained for 100K steps with an optimizer of *AdamW*, a learning rate $1e-5$, a warmup of 5K steps and a batch size of 512. The temperature τ of the contrastive objective function is set as 0.1. The balancing terms λ_i of the training objective are 0.5, 1, 1, and 0.2, respectively. Our model converges in about three hours on four 12 GB 2080ti GPUs, whereas conventional pre-training models like AnchiBERT (Tian et al., 2021) typically take several days on the same setup, depending on corpus size.

5.2. Baselines

We have categorized our baselines into three types for comparison: (1) Modern Chinese pre-trained models trained on the same ancient Chinese corpus as ours, (2) Cross-lingual pre-trained models utilizing the same parallel corpus, and (3) Ancient Chinese pre-trained models trained on a relatively extensive corpus.

³ It is available at <https://github.com/NiuTrans/Classical-Modern>, which is maintained by NLP Lab at Northeastern University and NiuTrans Research.

Table 1

Training corpus size for different models.

Model	Number of Training Tokens
AnchiBERT	39.5 M
SikuRoBERTa	536.1 M
GujiRoBERTa	925.3 M
Our model	17.5 M (44.8 M)

5.2.1. Modern Chinese PLMs

- **BERT-Base-Chinese** (Devlin et al., 2019): It is a BERT model based on simplified Chinese and traditional Chinese. There exist two distinct pre-training tasks, namely the Masked Language Model (MLM) and Next Sentence Prediction (NSP).
- **ERNIE** (Zhang et al., 2019): It aims to improve BERT's masking process by introducing entity-level and phrase-level masking. The named entities are masked by entity-level masking, whereas the consecutive words are masked by phrase-level masking.
- **BERT-wwm/RoBERTa-wwm** (Cui et al., 2021): Since the Chinese characters are structured differently from alphabetic languages, BERT-wwm/RoBERTa-wwm adopts the whole word masking (wwm) rather than splitting the word into small fragments.
- **ChineseBERT** (Sun et al., 2021): It incorporates both the pinyin and glyph aspects of Chinese characters into the BERT framework. Specifically, the pinyin embedding can characterize the phonetic aspects of Chinese characters, while the glyph embedding can capture the semantics of characters from the visual features.

5.2.2. Cross-lingual PLMs

- **XLM-R** (Conneau et al., 2020): It is a cross-lingual language model that utilizes a translation language model (TLM) built upon masked language modeling (MLM) to leverage cross-lingual parallel corpus datasets.

5.2.3. Ancient Chinese PLMs

- **AnchiBERT** (Tian et al., 2021): It is a pre-training model of ancient Chinese based on BERT-Base-Chinese, which is trained on ancient Chinese texts from 1000 B.C. to 200 B.C.
- **SikuRoBERTa** (Wang et al., 2022): The training of the model is conducted on a high-quality self-constructed corpus of ancient texts based on the *Si Ku Quan Shu*, initializing model parameters through RoBERTa-zh.
- **GujiBERT** (Wang, Liu, et al., 2023): It is trained on the ancient corpus of the *Daizhige*,⁴ which considers the coexistence of simplified and traditional characters.

We find that ERNIE 1.0 performs best compared to other versions (ERNIE 2.0 and ERNIE 3.0), so ERNIE 1.0 is used as the baseline. BERT-wwm and RoBERTa-wwm take into account the phrase structure of Chinese characters, while ChineseBERT takes into account the glyph and pinyin information. Both of the aforementioned baselines consider the characteristics of the Chinese. XLM-R provides an effective approach to leverage parallel corpora. To make fair comparisons, we use RoBERTa-wwm as the baseline instead of BERT-wwm. The models BERT-Base-Chinese*, RoBERTa-wwm*, ChineseBERT*, and XLM-R* are all fine-tuned on the same corpus as our model. AnchiBERT, SikuRoBERTa and GujiRoBERTa are all pre-trained models suitable for the ancient Chinese domain, which differ in the scope of their training corpus as shown in Table 1. In particular, the number of training tokens of the training parallel corpus for our model is 44.8M, of which ancient Chinese characters account for 17.5M.

Table 2Performances of different PLMs on *EvaHan2022* dataset. Results on test sets are reported for comparison (%).

Model	<i>EvaHan-WS</i>			<i>EvaHan-POS</i>		
	Precision	Recall	F1	Precision	Recall	F1
BERT-Base-Chinese*	93.67	93.76	93.71	86.99	87.26	87.13
ERNIE 1.0*	93.69	93.16	93.42	87.14	86.65	86.89
RoBERTa-wwm*	93.61	94.33	93.97	86.68	85.35	87.01
ChineseBERT*	93.85	94.23	94.04	87.05	87.41	87.23
XLM-R*	94.08	93.26	93.67	87.38	86.63	87.00
AnchiBERT	93.83	94.21	94.02	87.48	87.84	87.66
SikuRoBERTa	93.45	93.14	93.29	85.76	85.47	85.62
GujiRoBERTa	93.48	94.99	94.23	86.41	87.81	87.10
Our model	94.32	93.92	94.32	87.45	88.07	87.76

5.3. Main results

In this section, we perform comprehensive experiments on six ancient Chinese NLP tasks. For evaluation, models are fine-tuned separately on task-specific datasets.

5.3.1. Ancient Chinese word segmentation and POS tagging

It is a joint task, which aims to divide ancient Chinese text into words and then label the part-of-speech of each word. *EvaHan2022*⁵ datasets are used for this task, which is the first international evaluation dedicated to the evaluation of ancient Chinese word segmentation (*EvaHan-WS*) and part-of-speech tagging (*EvaHan-POS*). It consists of 7K/1K/0.9K sentences for training/dev/test. A mainstream sequence tagging model is utilized to evaluate this task, i.e., PLMs + Bi-LSTM + CRF.

As depicted in Table 2, our CCDM significantly outperforms the RoBERTa-wwm in terms of F1 scores for both tasks, while showing competitive results compared with the ancient Chinese pre-trained models. The results prove our model's capability in capturing syntactic variations between ancient and modern Chinese through a limited amount of parallel corpus, achieving the same effect as fine-tuning PLMs with a relatively large corpus.

5.3.2. Named entity recognition

The objective of this task is to recognize named entities within a piece of text. We use the *C-CLUE*⁶ NER dataset for this task, with 1.9K/0.2K/0.2K instances for training/dev/test. *C-CLUE* is a classical Chinese language understanding evaluation benchmark based on a crowd-sourcing annotation system, which is provided by Tianjin University. There are nearly 20000 entities obtained from the system annotation results, which are divided into 6 types of entities: person (PER), organization (ORG), etc. As shown in Table 3, our proposed model exhibits superior performance compared to all baseline models, as measured by F1 scores. We believe the reason is that our CCDM is able to capture the underlying finer-grained syntactic information, which is beneficial for token-level language understanding tasks.

5.3.3. Relation extraction

This task is asked to identify the relation between two named entities within a piece of text. We use the RE dataset in *C-CLUE*⁷ for this task, with 3K/0.9K/0.4K instances for training/dev/test. There are nearly 4000 relations obtained from the crowd-sourcing annotation system, which are divided into 7 types of relations: PER-PER, PER-ORG, PER-LOC, PER-POS, etc. As shown in Table 4, our model outperforms all the baselines in terms of precision and Micro-F1. The results demonstrate that our model has a better understanding of both token-level and sentence-level information, which is crucial for relation extraction.

⁵ <https://circse.github.io/LT4HALA/2022/EvaHan>

⁶ <https://github.com/jizijing/C-CLUE/tree/main>

⁷ https://github.com/jizijing/C-CLUE/tree/main/sequence_labeling_data_guwen

⁴ <https://github.com/garychowcmu/daizhige20>

Table 3

Performances of different PLMs on *C-CLUE* NER dataset. Results on the test set are reported for comparison (%).

Model	<i>C-CLUE NER</i>		
	Precision	Recall	F1
BERT-Base-Chinese*	56.85	74.87	64.63
ERNIE 1.0*	56.65	76.53	65.11
RoBERTa-wwm*	58.49	75.00	65.72
ChineseBERT*	58.35	76.19	66.09
XML-R*	59.79	70.12	64.54
AnchiBERT	60.01	73.94	66.25
SikuRoBERTa	59.23	75.53	66.39
GujiRoBERTa	60.58	72.55	66.02
Our model	60.91	75.22	67.31

Table 4

Performances of different PLMs on *C-CLUE* RE dataset. Results on the test set are reported for comparison (%).

Model	<i>C-CLUE RE</i>		
	Precision	Recall	F1
BERT-Base-Chinese*	44.07	50.11	46.90
ERNIE 1.0*	44.05	51.02	47.28
RoBERTa-wwm*	46.38	50.94	48.55
ChineseBERT*	47.17	51.65	49.31
XML-R*	46.63	50.58	48.52
AnchiBERT	49.35	51.08	50.20
SikuRoBERTa	49.15	50.85	49.99
GujiRoBERTa	48.28	51.08	49.64
Our model	49.59	50.92	50.25

Table 5

Performances of different PLMs on *Daizhige* dataset. Results of accuracy on dev and test sets are reported for comparison (%).

Model	<i>Daizhige</i>	
	Dev	Test
BERT-Base-Chinese*	74.30	76.40
ERNIE 1.0*	74.25	76.30
RoBERTa-wwm*	73.75	75.45
ChineseBERT*	73.25	73.85
XML-R*	72.80	73.55
AnchiBERT	72.55	74.95
SikuRoBERTa	72.70	74.25
GujiRoBERTa	77.10	77.05
Our model	<u>75.30</u>	<u>76.80</u>

5.3.4. Text classification

This task aims to categorize a piece of ancient text into one of the specified classes. We use a subset of *Daizhige*⁸ for this task, with 10K/2K/2K instances respectively for training/dev/test. It contains a complete collection of ancient documents in 10 domains, including poetry, medicine, etc.

As shown in *Table 5*, our model outperforms all the baselines except GujiRoBERTa. A key reason for this is that GujiRoBERTa was fine-tuned on the entire *Daizhige* corpus, while our training corpus covers less than 2% of it. It is important to clarify that the task itself is a supervised text classification challenge based on the *Daizhige* corpus, while both GujiRoBERTa and our model were initially pre-trained using unsupervised learning methods, ensuring there is no label leakage. However, the extensive coverage of the *Daizhige* corpus during GujiRoBERTa's pre-training phase could explain its enhanced performance, as it has been more thoroughly adapted to the specific language and style of these texts. Despite this, our model still achieved the second-best performance. This indicates that our model can acquire in-depth language knowledge with a relatively small corpus via contrastive learning of ancient and modern parallel corpora.

⁸ <https://github.com/garychowcmu/daizhige20>

Table 6

Performances of different models on *FSPC* datasets. Results of accuracy (Acc) on dev and test sets are reported for comparison (%).

Model	<i>FSPC</i>	
	Dev	Test
BERT-Base-Chinese*	59.50	59.75
ERNIE 1.0*	59.25	60.15
RoBERTa-wwm*	57.95	58.85
ChineseBERT*	59.30	59.15
XML-R*	59.60	59.85
AnchiBERT	61.15	60.20
SikuRoBERTa	60.25	61.35
GujiRoBERTa	59.40	61.60
Our model	61.20	<u>61.45</u>

Table 7

Performances of different models on MRC datasets. Results of accuracy on dev and test sets are reported for comparison (%).

Model		
	Dev	Test
BERT-Base-Chinese*	34.00	27.70
ERNIE 1.0*	29.25	28.64
RoBERTa-wwm*	36.50	34.51
ChineseBERT*	29.04	27.15
XML-R*	36.50	33.58
AnchiBERT	39.00	38.03
SikuRoBERTa	35.75	35.92
GujiRoBERTa	41.50	38.26
Our model	42.00	40.61

5.3.5. Poem sentiment analysis

The PSA task aims to calculate the corresponding sentiment labels of poetry. We conduct experiments on a publicly released labeled dataset - *FSPC*,⁹ with 4K/0.5K/0.5K instances respectively for training/dev/test. *FSPC* is a manually-labeled Fine-grained Sentiment Poetry Corpus provided by Tsinghua University which contains 5,000 pieces of poems. It is divided into five categories: negative, implicit negative, neutral, implicit positive, and positive. Results are shown in *Table 6*, where we can see that our model outperforms the majority of baselines on dev and test sets. Although the format of poetry is different from that of prose, they both belong to ancient Chinese, whose word semantics are similar. This task implies that our model has some generalization ability on comprehension tasks related to ancient Chinese.

5.3.6. Machine reading comprehension

Given a piece of text context *C* and question *Q*, the goal of the model is to give the correct answer regarding question *Q* from several choices. We experiment with an available dataset for the ancient Chinese machine reading comprehension (MRC) task,¹⁰ with 3.3K/0.4K/0.4K instances for training/dev/test. The dataset contains both ancient and modern Chinese texts, see the supplementary material for a detailed example. As shown in *Table 7*, our model achieves optimal results. In particular, our model shows a significant improvement compared to ancient Chinese pre-trained models. The reason is that these ancient Chinese PLMs pay more attention to the ancient Chinese understanding and yield a certain deficiency in modern Chinese understanding. In contrast, our model is able to capture the gap between modern and ancient Chinese without losing the comprehension of modern Chinese in the meantime.

5.3.7. Comparative performance discussion

In this study, we have categorized models into three types: modern Chinese PLMs, cross-lingual PLMs, and ancient Chinese PLMs. It is

⁹ The data is available at <https://github.com/THUNLP-AIPoet/Datasets/tree/master/FSPC>.

¹⁰ It is available at <https://github.com/Ethan-yt/CCLUE/tree/main/data/mrc>. We redivide the dataset since the test set is not standardized. The final number of available instances is 4.1K.

Table 8

Statistical significance of task performance improvements.

Task	WS	POS	NER	RE	TC	PSA	MRC
P-Value	0.0026	0.0061	0.0003	0.0120	0.0106	0.0056	0.0021

Table 9

Ablation study results of different PLMs on C-CLUE NER dataset. Results on the test set are reported for comparison (%).

Model	Precision	Recall	F1
CCDM	60.91	75.22	67.31
w/o WPL	60.57	75.10	67.02
w/o CCL	59.62	74.84	66.37
w/o CRL	60.18	74.99	66.77
RoBERTa-wwm	57.84	74.40	65.08

evident that our model outperforms the modern Chinese PLMs and cross-lingual PLMs, and achieves competitive results compared to those ancient Chinese PLMs trained on a substantial corpus of classical literature. For the marginal improvement, we conduct t-tests for significance testing, as shown in Table 8. The p-values for all tasks are below 0.05, indicating that our model shows significant improvements.

This performance enhancement can be attributed primarily to two factors: (1) The new decoupling strategy is able to distinguish fine-grained syntactic gap information, which is beneficial for the pre-trained model to distinguish ancient Chinese from modern Chinese. (2) Our proposed cross-temporal contrastive learning can efficiently exploit the limited parallel corpus, mining the syntactic and semantic information underlying the corpus.

5.4. Analysis

5.4.1. Ablation studies

We conduct ablation studies of CCDM on the C-CLUE NER dataset by removing word position loss (represented by CCDM w/o WPL), cross-temporal contrastive loss (represented by CCDM w/o CCL), and cross-temporal reconstruction loss (represented by CCDM w/o CRL), to investigate their contributions respectively.

As shown in Table 9, our model achieves the strongest performance among all variants, illustrating the importance of all proposed components. In particular, we observe a significant decline in the performance of the variant CCDM w/o CCL compared to the variant CCDM w/o CRL. We argue that although CRL improves the performance by exploiting cross-temporal reconstruction as a strong constraint, it rarely pays attention to the representation space gap between ancient and modern Chinese, while CCL compensates for this deficiency. In addition, it can be seen that all the variants perform significantly better than the initial pre-trained model RoBERTa-wwm, indicating that there is a distinction between the semantics and syntax of ancient Chinese and modern Chinese, and that our CCDM is well capable of capturing it.

5.4.2. Does the disentangled representation work?

Our newly proposed decoupling strategy VG²VAE aims to learn the syntactic differences and similarities between ancient Chinese and modern Chinese. The previous decoupling strategies only decoupled syntax and semantics without considering the syntactic similarity, which exists between ancient and modern Chinese. Therefore, the previous decoupling strategy is not suitable for our task.

To verify the effectiveness of our proposed decoupling strategy VG²VAE compared with the previous strategy VGVAE, we conduct an experiment where we replace the new decoupling strategy with the previous one and retrain the model and evaluate its performance on the NER dataset. The results of this experiment are shown in Fig. 5. we replace the new decoupling strategy with the previous decoupling strategy, retrain the model, and carry out experiments on the NER dataset. Results are shown in Fig. 5 where our proposed decoupling

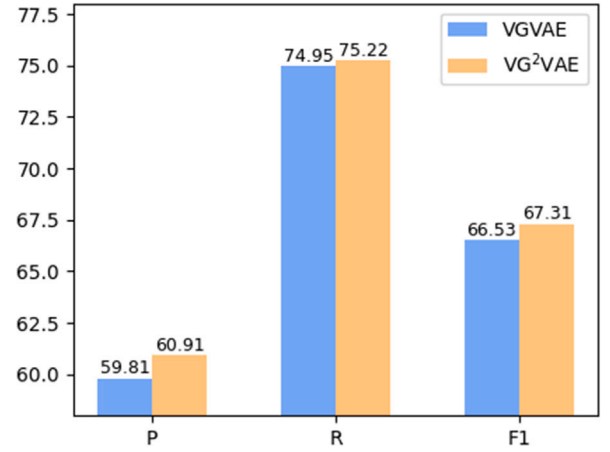


Fig. 5. Performances of models with different decoupling strategies on C-CLUE NER datasets. Results of precision (P), recall (R), and F1 on test sets are reported for comparison (%).

strategy is superior to the previous decoupling strategy. This is because our model allows PLM to learn the syntax gap between ancient and modern Chinese at a finer granularity.

For analyzing the actual effect of our decoupling strategy, we visualize the hidden representations of the last layer of PLMs and disentangled semantic and syntactic representations of VG²VAE from 1,000 parallel sentence pairs in Fig. 6. The visualizations of Fig. 6(a) and (b) show that our model can learn the difference between ancient Chinese and modern Chinese well compared to the initial model RoBERTa-wwm. Moreover, it is evident that disentangled semantic and same-syntactic representations learned by VG²VAE make parallel sentences closer to one another in space (Fig. 6(c), (d)). The disentangled different-syntactic representations learned by VG²VAE are far away from one another in space (Fig. 6(e)). The visualizations demonstrate that VG²VAE is able to efficiently disassociate semantics from syntax and different-syntax from same-syntax.

5.4.3. How does CCDM perform with different training corpus sizes?

We train the CCDM in different training corpus size settings by randomly sampling 10% to 80% from the full training set to construct a low-resource training set. We perform experiments on the C-CLUE NER dataset and present the results in Fig. 7. The training percentage of 0 represents the initial model RoBERTa-wwm. It is evident from Fig. 7 that there is a noticeable decline in model performance when the training percentage is set to 10%. We guess that it might be due to the overfitting of model parameters caused by insufficient training data. With the increase of the training percent to 40, our model remarkably improves in performance compared to the initial pre-training model RoBERTa-wwm. It is noteworthy that the training corpus, in this case, is also considerably less than the corpus utilized in other ancient Chinese pre-training models. This further demonstrates that our model can distinguish ancient and modern Chinese with less corpus, achieving the knowledge transfer from modern Chinese to ancient Chinese.

5.4.4. Case studies

To more granularly validate the model's performance in syntactic and semantic decoupling, we compared three of the most optimal classical Chinese pre-trained models: AnchiBERT, SikuRoBERTa and GujiRoBERTa. Specifically, we constructed a Milvus vector database based on the ancient-modern translation corpus from the "Old Book of Tang" (旧唐书) for similarity retrieval, aiming to analyze the models' decoupling capabilities in syntax and semantics. In this experiment, the classical Chinese pre-training models generate sentence vectors

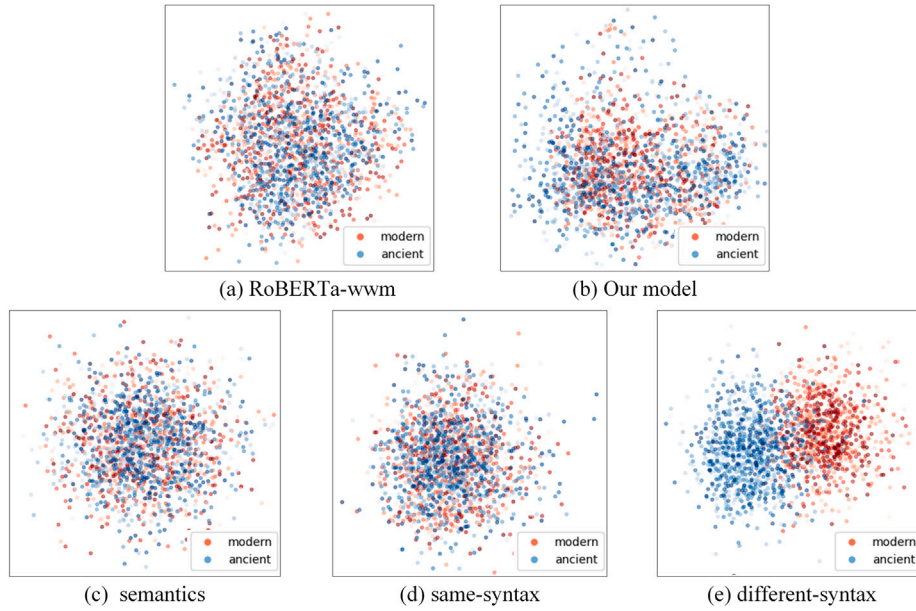


Fig. 6. Visualization of hidden representations of the last layer of PLMs (a–b) and disentangled semantic and syntactic representations of VG²VAE (c–e).

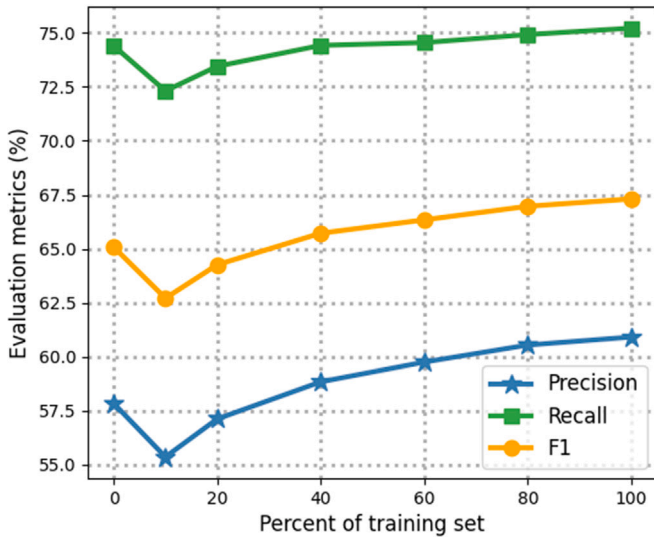


Fig. 7. Performance comparison in different training corpus size scenarios.

for matching and retrieval, whereas our model leverages the decoupling of semantics and syntax to provide multidimensional retrieval functionalities—encompassing both semantic (Sem) and syntactic (Syn) searches. These retrievals are implemented by concatenating the PLM’s output with the decoupled semantic and syntactic vectors.

In an ideal scenario, it is expected to be able to retrieve the corresponding translated sentences (R). As shown in Fig. 8, for each query, we can retrieve the most semantically and syntactically similar sentences from the vector databases. Classical Chinese pre-training models like AnchiBERT tend to return sentences in classical Chinese, often disregarding the syntactic differences between classical and modern Chinese. As a result, they frequently fail to retrieve the correct translated sentences. In contrast, our model leverages decoupled semantic information to retrieve sentences from multiple dimensions, extending beyond just classical texts.

Moreover, our model is not limited to semantic matching but also enables syntactic vector retrieval, identifying sentences that are syntactically most similar to the query. Thus, users engaged in syntactic

analysis can find syntactically similar sentences, providing richer reference materials and enhancing their ability to parse and translate classical Chinese texts.

5.4.5. Error analysis

Error analysis is essential for identifying potential improvements in our model. Through case studies, we have selected a classic example to illustrate two key areas for enhancement.

As shown in Fig. 9, at the semantic level, our model, similar to the classical Chinese pre-training model, fails to accurately retrieve the modern Chinese translation. This issue arises because classical Chinese frequently employs metaphorical expressions and interchangeable characters (as highlighted in red in the figure), which can lead to incorrect interpretation of the text’s meaning. For instance, “枳棘” (zhǐ jí) originally denotes nouns representing thorny plants, but here it is used metaphorically to indicate creating obstacles for someone. Similarly, “涂” (tú), a homophone for “途” (tú), meaning road or path. Therefore, incorporating more granular word-level semantic knowledge of both classical and modern Chinese could significantly enhance the model’s performance.

On the syntactic level, it is evident that only a small portion of the syntax tree of the sentences retrieved by our model resembles that of the query. This indicates that our model still has limitations when dealing with long texts and complex syntactic structures. Thus, in future research, it may be beneficial to incorporate explicit syntactic tree structures. By integrating these structures, we can more effectively analyze and understand complex syntactic relationships, thus enhancing the model’s retrieval capabilities and interpretability.

In conclusion, these findings suggest that addressing both semantic and syntactic challenges is crucial for advancing our model’s capabilities. Enhanced handling of metaphorical language and interchangeable characters, along with the adoption of syntactic tree structures, represent promising directions for future research.

6. Limitations

Although our model can effectively utilize information from parallel corpora to understand low-resource languages, the limitations brought by parallel corpora must be acknowledged, especially when generalizing to other languages.

Most Similar Sentences	Dependency Trees
Query: 度, 贞元五年进士擢第, 登宏辞科。① Ref: 裴度于贞元五年考中进士, 中选宏辞科。① Anchi: 禹锡贞元九年擢进士第, 又登宏辞科。② Siku: 禹锡贞元九年擢进士第, 又登宏辞科。② Guji: 禹锡贞元九年擢进士第, 又登宏辞科。② Ours { Sem: 裴度于贞元五年考中进士, 中选宏辞科。① Syn: 禹锡贞元九年擢进士第, 又登宏辞科。② Eng: ① Liu Yuxi passed the imperial examination and became a jinshi in the ninth year of Zhenyuan, and later succeeded in the Hongci Examination as well. ② Pei Du passed the imperial examination and became a jinshi in the fifth year of Zhenyuan, subsequently succeeding in the Hongci Examination.	Query Root HED 擢第 SBV ADV VOB WP COO 度 贞元 进士 , 登 五年 VOB 宏辞科 Ours syntactic Root HED 擢 SBV ADV VOB WP COO 禹锡 贞元 第 , 登 九年 ATT 进士 ADV VOB 又 宏辞科
Query: 主人赂而谢之, 方肯携蛇簠而去。① Ref: 店东贿赂、道歉, 小使们才肯将蛇箱带走。① Anchi: 义方解所乘马与之, 不告姓名而去。② Siku: 一行承其言而趋入, 稽首请法, 尽受其术焉, 而门前水果却西流。③ Guji: 一行承其言而趋入, 稽首请法, 尽受其术焉, 而门前水果却西流。③ Ours { Sem: 店东贿赂、道歉, 小使们才肯将蛇箱带走。① Syn: 义方解所乘马与之, 不告姓名而去。② Eng: ① Only after the shop owner bribed and apologized did the servants agree to take away the snake box. ② Wang Yifang dismounted his horse and handed it over to him, then left without telling him his name. ③ Following his instructions, Yi xing hurriedly entered the yard, kowtowed and begged to be taught the divination. After learning all the skills from him, the stream in front of the courtyard indeed flowed westward.	Query Root HED 赂 SBV COO WP COO 主人 谢 , 携 ADV VOB ADV VOB COO 而 之 方肯蛇簠去 ADV 而 Ours syntactic Root HED 解 SBV VOB COO WP COO 义方 马 与 , 不告 ATT VOB VOB COO 所乘 之 姓名 去 ADV 而

Fig. 8. Examples of most similar sentences to particular query sentences (Q), as retrieved by four different models: AnchiBERT (Anchi), SikuRoBERTa (Siku), GujiRoBERTa (Guji), and our model. R represents the modern Chinese translation reference sentences. Human-annotated English translations and dependency trees are provided for easier reading and understanding.

The effectiveness of the CCDM depends heavily on the availability of parallel corpora. Such corpora may be scarce due to historical and practical constraints, limiting the scalability and generalizability of our model. Although our model is designed to efficiently utilize minimal parallel data (as demonstrated in Section 5.4.3), this requirement can still be a bottleneck. Constructing parallel corpora is often time-consuming and resource-intensive, requiring extensive expertise.

Fortunately, the emergence of advanced large language models (LLMs), such as ChatGPT and LLaMA, offers promising pathways for generating high-quality parallel corpora. These models can significantly support and enhance resources for low-resource languages. Therefore, exploring methodologies for generating parallel corpora using LLMs will be a key focus of our future work.

7. Conclusion and future work

In this paper, we propose a novel language pre-training framework for ancient Chinese understanding based on Cross-temporal Contrastive

Disentangling Model (CCDM). This model is able to bridge modern Chinese and ancient Chinese through a limited parallel corpus, since it captures their semantic and syntactic changes. We conduct extensive experiments on six ancient Chinese understanding tasks and demonstrate the effectiveness of the proposed framework. Additionally, the framework shows potential applicability to other languages characterized by the evolutionary nature, where syntax and semantics have undergone some changes.

Despite the model's proficiency in discerning the disparities between ancient and modern language forms, substantial opportunities for advancement remain. Leveraging LLMs could help overcome the challenges associated with the limited availability of parallel corpora, thereby potentially enhancing the model's performance significantly. Additionally, the incorporation of knowledge graphs could address intricate semantic challenges, while the integration of explicit syntactic trees would facilitate deeper understanding and analysis of syntactic structures. Moving forward, we aim to explore these improvements more deeply, further advancing our model's capabilities.

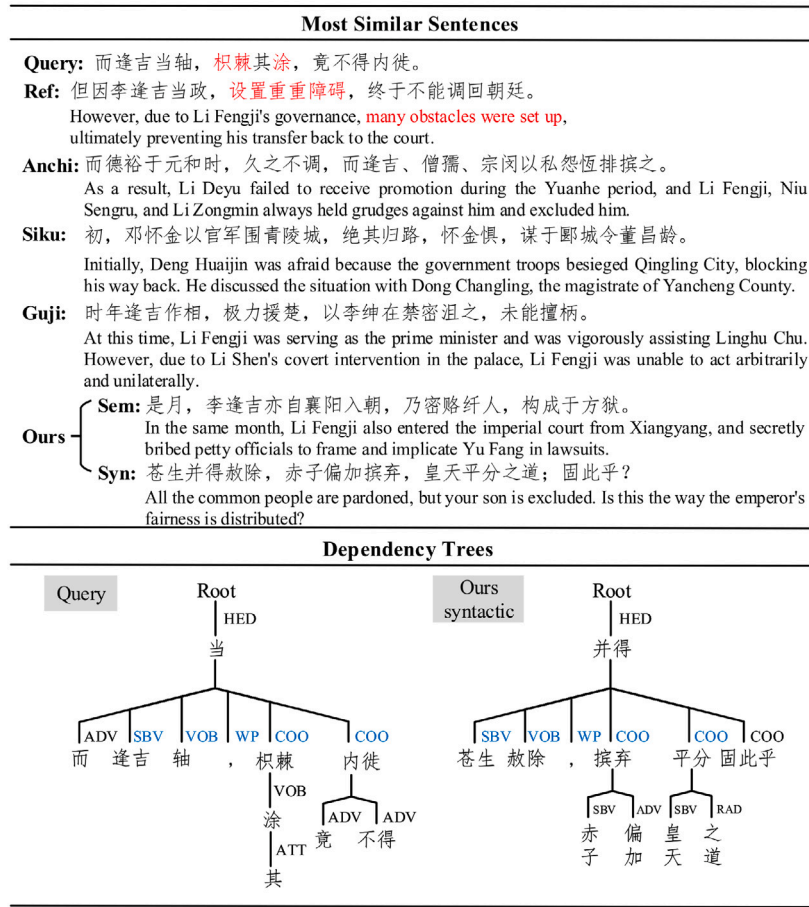


Fig. 9. Examples of most similar sentences to particular query sentences (Q), as retrieved by four different models: AnchiBERT (Anchi), SikuRoBERTa (Siku), GujiRoBERTa (Guji), and our model. R represents the modern Chinese translation reference sentences. Human-annotated English translations and dependency trees are provided for easier reading and understanding.

CRedit authorship contribution statement

Yuting Wei: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Yangfu Zhu:** Writing – review & editing, Conceptualization. **Ting Bai:** Writing – review & editing, Methodology. **Bin Wu:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. An example of ancient Chinese MRC

The data for ancient Chinese MRC are obtained from the Chinese examination papers for primary and secondary schools. Given an

Content	鱼我所欲也 --《孟子》 鱼,我所欲也;熊掌,亦我所欲也。[...] <i>Fish Is What I Desire</i> by Mencius Fish is what I desire, and so are bear's paws.
Question	对选文内容的理解、分析、概括有误的一项是 Which one is wrong in understanding, analyzing or summarizing the content of the selection?
Choices	① 本文开篇运用了比喻论证的方法,以鱼和熊掌为喻。[...] This content opens with a metaphorical argument, using a fish and a bear's paw as metaphors. ② 面对生死抉择的时候,那些“舍生取义”者表现的是“所欲有甚于生者,故不为苟得”[...] When faced with the choice of life and death, those who "choose the righteousness and forsake the life" behave as "not stick to life discredibly if something more precious is to be desired". ③ ...

Fig. A.10. An example of ancient Chinese MRC. The words in red are ancient Chinese, while the others are modern Chinese.

ancient text content and a question, the reader is tasked with selecting the accurate answer from a provided list of options. As shown in Fig. A.10, it can be seen that the content is ancient Chinese, while the choices are a mixture of ancient and modern Chinese. Hence, this task requires PLMs to have a better understanding of both ancient and modern Chinese.

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